Classification

A requirement from the Hospital, Management asked us to create a predictive model which will predict the Chronic Kidney Disease (CKD) based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement

The objective of this project is to develop a **machine learning classification model** that can predict whether a patient has **Chronic Kidney Disease (CKD)** or not, based on various clinical and physiological parameters provided in the dataset.

This is a **binary classification problem**:

- Output:
  - **Yes (CKD present)**
  - **No (CKD not present)**

2.) Tell basic info about the dataset (Total number of rows, columns)

- **Total Rows (Records):** 399
- **Total Columns (Features):** 28

Types of Features:

- **Numerical features:** age, bp (blood pressure), etc.
- **Categorical features:** rbc, pc, appet, etc.
- **Target column:** classification (yes/no)

2.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

To prepare the dataset for modeling, the following preprocessing steps were applied:

a) **Encoding Categorical Data (Nominal Data)**

Since ML models work with numbers:

- Converted categorical values like:
  - yes/no → 1/0
  - normal/abnormal → 1/0

b) Feature Scaling

- Applied **Standardization** using:
  - StandardScaler

- This ensures all features are on the same scale (important for SVM, KNN).

c)Splitting Dataset

- Training set: 80%
- Testing set: 20%

4.) Develop a good model with good evaluation metric. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Multiple models were trained and evaluated:

Algorithms Used:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Naïve Bayes
4. Support Vector Machine (SVM)
5. Random Forest
6. Decision Tree

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

5.) All the research values of each algorithm should be documented. (You can make tabulation or screenshot of the results.)

Algorithm	**Confusion Matrix**	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC**
Logistic Regression	[[45 0] [1 74]]	**0.99**	0.98 **1.0**	1.00 **0.99**	0.99 **0.99**	1.0
KNN	[[45 0] [6 69]]	0.95	0.88 **1.00**	1.00 **0.92**	0.94 **0.96**	0.99
Naïve Byes	[[45 0] [3 72]]	0.97	0.94 **1.00**	1.00 **0.96**	0.97 **0.98**	1.0
SVM	[[45 0] [2 73]]	0.98	0.96 **1.00**	1.00 **0.97**	0.98 **0.99**	0.99
DT Classification	[[45 0] [5 70]]	0.96	0.90 **1.00**	1.00 **0.93**	0.95 **0.97**	0.96
Random Forest	[[44 1] [1 74]]	0.98	0.98 **0.99**	0.98 **0.99**	0.98 **0.99**	0.99

*Note in the above table for column –

Precision, Recall, F1score **RED** indicates 0 and **GREEN** Indicated 1

6.) Mention your final model, justify why u have chosen the same.

Final Model:

Logistic Regression

Accuracy: 0.99

Why this model was selected

Logistic Regression was chosen because:

1. Accuracy is 0.99 among all models.
2. F1-score = 0.99, indicating excellent balance between precision and recall
3. ROC-AUC = 1.0, showing perfect class separability
4. In confusion matrix -False Negatives (FN) ≈ 0 or very minimal which means almost no CKD patients are missed
5. Recall for CKD class ≈ **1.00. The model correctly identifies almost all CKD patients**